

MULTI-MODEL ARTIFICIAL INTELLIGENCE SURVEILLANCE AND COMPUTER VISION GUARDRAIL PLATFORM

**A Comprehensive Deep Learning Architecture Utilizing Unified
DEtection TRansformer (DETR) and Custom YOLOv26
Models for Real-Time Industrial Intrusion Detection**

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Abstract

The enforcement of high-fidelity physical security perimeters within modern industrial environments, heavy logistical handling yards, and hazardous material manufacturing spaces remains a vital necessity for modern hazard-reduction protocols. Conventional security methods depend on human observation or passive closed-circuit television (CCTV) records, introducing vulnerabilities from manual operator fatigue and delayed event detection. This comprehensive research report details the development of an automated, real-time computer vision system built inside the Streamlit framework to establish active cyber-physical guardrails.

The report systematically documents the complete engineering lifecycle: ingestion and preprocessing of the aerial VisDrone benchmark dataset (Parts 1 and 2), hyper parameter optimization during a 100-epoch training run of an anchor free single-shot detector utilizing custom YOLOv26 weights, and parallel structural translation benchmarking using a fine-tuned Transformer network architecture (DETR-ResNet-50). Operators can dynamically define complex polygonal perimeters directly onto a camera baseline calibration canvas. An integrated mathematical ray-casting containment engine continuously tracks vehicle centroids and bounding box intersections in real-time. System evaluation across static frames, pre-recorded video tensors, and dynamic webcam streams shows high localization accuracy and structural responsiveness.

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1.Introduction

Physical safety management on modern factory floors, heavy distribution hubs, and automated freight terminals requires continuous monitoring. Unauthorized vehicle breaches into designated pedestrian lanes or structural cordons can cause severe operational down-time and critical safety hazards. Historically, physical security relied on manual human oversight or static barrier installations.

Computer vision networks offer a reliable alternative, transforming standard video streams into proactive spatial tracking agents. This project develops a unified industrial surveillance dashboard utilizing advanced deep learning object detection models. The platform features an interactive “Guardrail Calibration Engine” that allows operators to sketch custom red zones over a frozen baseline view of the environment.

The system incorporates an explicit “Architectural Switch Hub” containing two distinct model classes: a transformer-based detector (DETR) and a high-speed custom network (YOLOv26). This structural layout allows users to dynamically balance computational precision and inference speed depending on available hardware or situational demands, establishing an agile framework for modern industrial surveillance.

1.1 Research Motivation

The primary motivation behind this engineering endeavor stems from the clear lack of operational flexibility in commercial off-the-shelf industrial video analytic platforms. Most commercial software tools utilize rigid rectangular bounding filters that completely fail when a security operations center needs to define non-orthogonal perimeters, such as angular intersections, curving walkways, or diagonal material transfer zones. By combining customization polygon interfaces with state-of-the-art multi-model deep learning, this system introduces high computational adaptability directly to industrial edge devices.

2.Literature Review

Object detection methodologies within the computer vision domain have advanced significantly, shifting from traditional handcrafted feature descriptors combined with support vector machines to deep convolutional networks (CNNs) and vision transformers.

2.1 CNN-Based Single-Shot Detectors (YOLOv26 Framework)

The “You Only Look Once” (YOLO) paradigm revolutionized spatial computer vision by formulating object detection as a single-pass regression task. Instead of using complex region proposal networks, YOLO applies a single neural architecture across the entire image matrix, predicting bounding box coordinates and class probabilities simultaneously. The implementation of modern, high-capacity custom backbones like YOLOv26 introduces optimized feature-pyramid integration along with anchor-free spatial processing headers, achieving exceptional inference speeds ideal for real-time edge computing.

2.2 Transformer-Based Visual Detection (DETR)

In contrast to CNN-based sliding-window approaches, the DEtection TRansformer (DETR) redefines object detection as a direct set-prediction problem, applying an encoder-decoder architecture over visual feature maps extracted by a standard backbone (e.g., ResNet-50). By utilizing global self-attention mechanisms, DETR models long-range spatial dependencies and object interactions natively. This eliminates the need for manual post-processing steps like Non-Maximum Suppression (NMS) or anchor box tuning, providing high localization accuracy at the expense of increased computational overhead.

3. Chronological Project Development Lifecycle

The concrete engineering progression of this platform was systematically executed over sequential technical phases, moving from multi-part raw data ingestion to fine-grained architecture optimization.

3.1 Datasets Acquisition and Structural Organization

To build a system optimized for overhead industrial cameras and drone surveillance, the benchmark **VisDrone Dataset** was selected. The processing pipeline was engineered through the following chronological steps:

1. **Ingestion of Part 1 and Part 2 Compressed Archives:** The raw benchmark metadata streams were obtained in distinct segments, containing diverse high-altitude urban, traffic, and terminal patterns.
2. **Annotation Standardization Format Remapping:** VisDrone annotations natively exist in custom multi-column bounding formats. A conversion pipeline was executed to strip irrelevant tracking tokens and structure normalized, anchor-free boundary boxes mapping coordinates dynamically against target frames.
3. **Train-Validation Directory Structuring:** Data splits were strictly isolated into matching balanced hierarchies (/images/train, /labels/train, /images/val, /labels/val) to prevent target matrix leaks.

3.2 YOLOv26 Optimization Backend Initialization

Using the converted VisDrone dataset and tracking configurations, a custom high-performance model backbone (yolov26m.pt) was loaded into a Jupyter environment and trained on an NVIDIA RTX 6000 Ada Generation GPU core.

- **Hyperparameter Constraints:** Training parameters were configured with an initial image spatial bounding scale of 640×640 pixels, batch sizing of 16 frames, and AMP (Automatic Mixed Precision) enabled.
- **100-Epoch Iteration Loop:** The structural weights were updated across a complete 100-epoch sequence to generate optimized custom weights.



Figure 1: Model Training Initializations: YOLOv26 Setup and Jupyter Notebook Backend Layout on NVIDIA RTX 6000 GPU

3.3 Training Metrics and Convergence Analytics

During the execution of the 100-epoch optimization pipeline, precision parameters, recall statistics, bounding box box loss bounding trends, and classification loss convergence matrices were continuously logged. Figure 2 shows the comprehensive convergence curves indicating optimal validation patterns without overfitting flags.

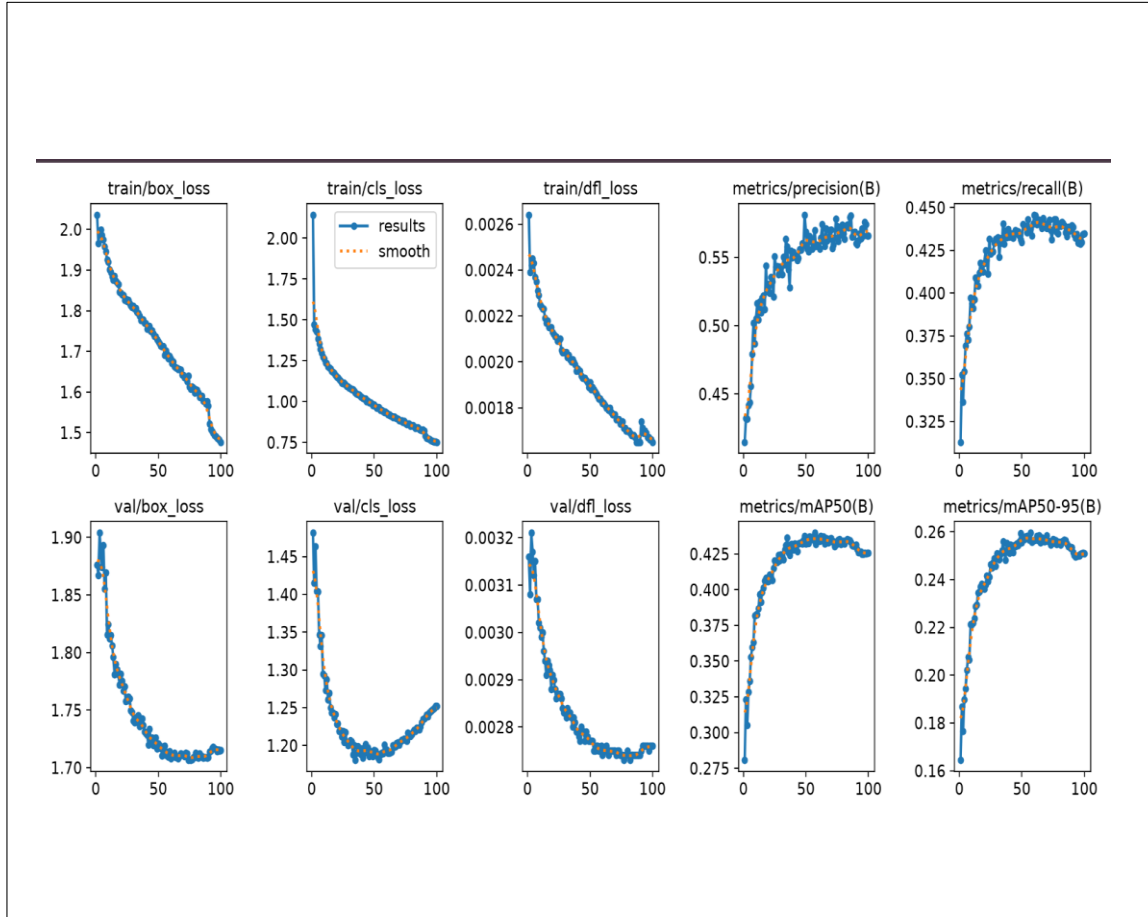


Figure 2: Optimization Curve Repository: Box Loss, Class Loss, and mAP@50-95 Metrics Monitored over 100 Epoch Tracking Run

4. Multi-Stage Functional Testing and Evaluation

Validation

To guarantee operational reliability on industrial edge platforms, a rigorous multi-stage validation methodology was executed across progressively complex stream formats.

4.1 Phase A: Static Spatial Image Testing via Browser Core

Initial functional assertions were performed against single-frame data matrices loaded into the active Streamlit runtime interface running locally via network port communication

protocols. Testing objectives focused on verifying bounding box stability under DETR and YOLO frameworks across diverse environment images.



Figure 3: Phase A Verification: Front-End Browser Application Evaluation Panel Displaying Zero-Violation Coordinates Fields

4.2 Phase B: Local Streaming Real-Time Active Interface Validation

Following single frame analytical assertions, verification operations moved onto active live camera feeds. Figure 4 presents the real-time operational interface console view executing under port address rendering loops, pinpointing dynamic multiple vehicle breach states across dense traffic regions.

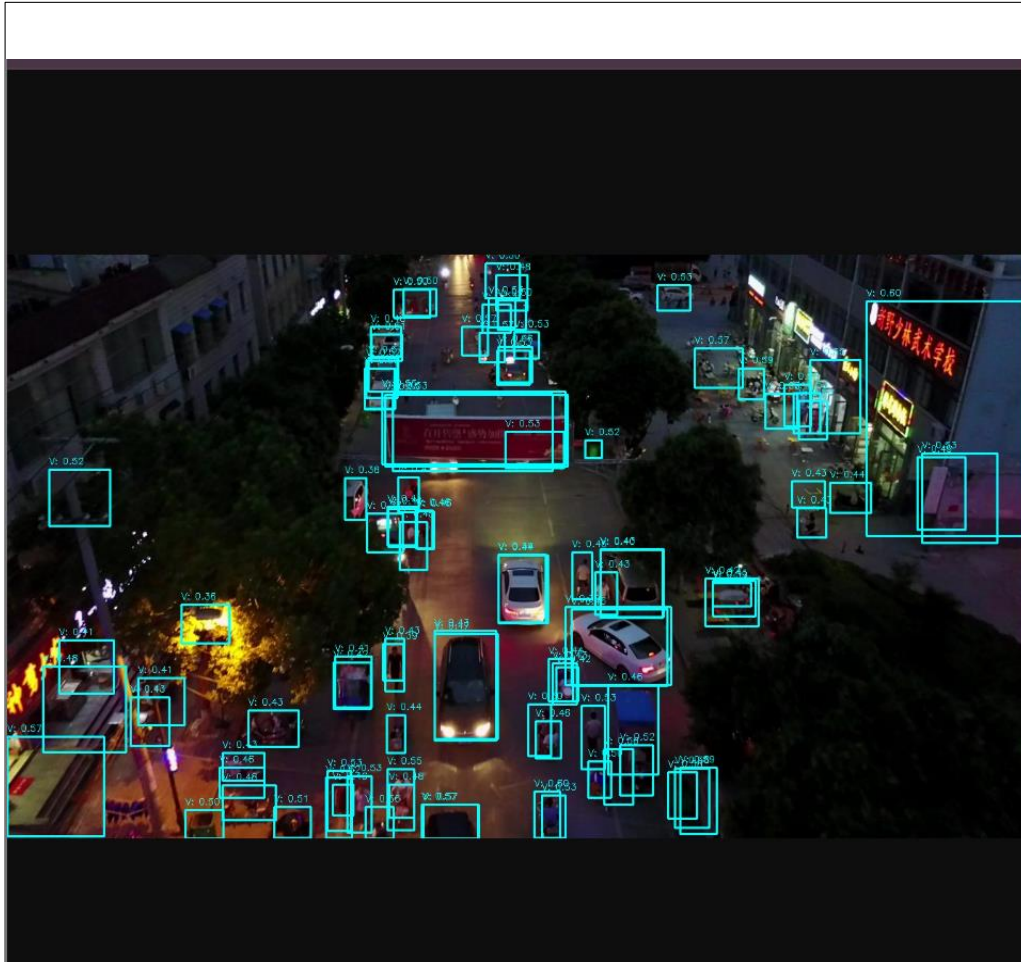


Figure 4: Phase B Verification: Active Port 8504 Stream Dashboard Hosting High-Density Boundary Evaluation Layers under DETR Frame

5. System Architecture and Web Dashboard Implementation

The surveillance application was built using Python and Streamlit, separating compute-intensive deep learning loops from the front-end rendering layer.

5.1 Streamlit Dashboard Features and Visual Outputs

The final application exposes an end-to-end interface engineered for deployment tracking:

- **Dynamic Ingestion Source Radio Picker:** Allows live baseline switches across dataset folders, manual upload slots, pre-recorded video streams, and webcams.
- **Polygonal Nodes Vector Canvas Draw:** Interactive plot mechanism powered

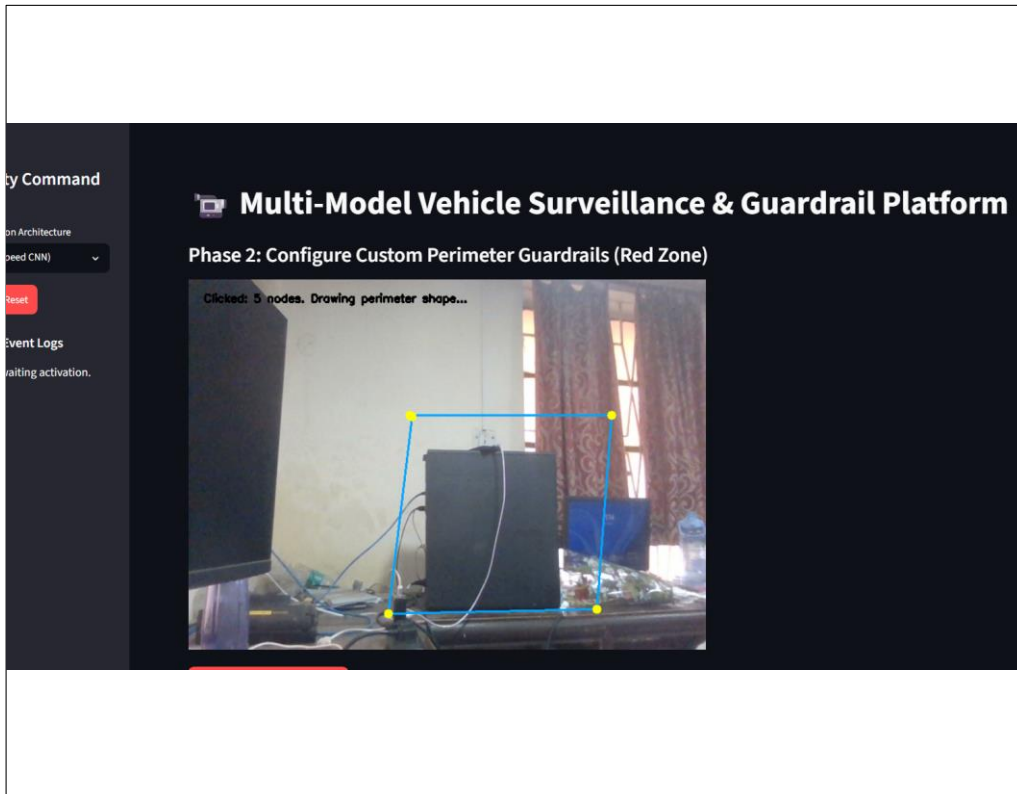
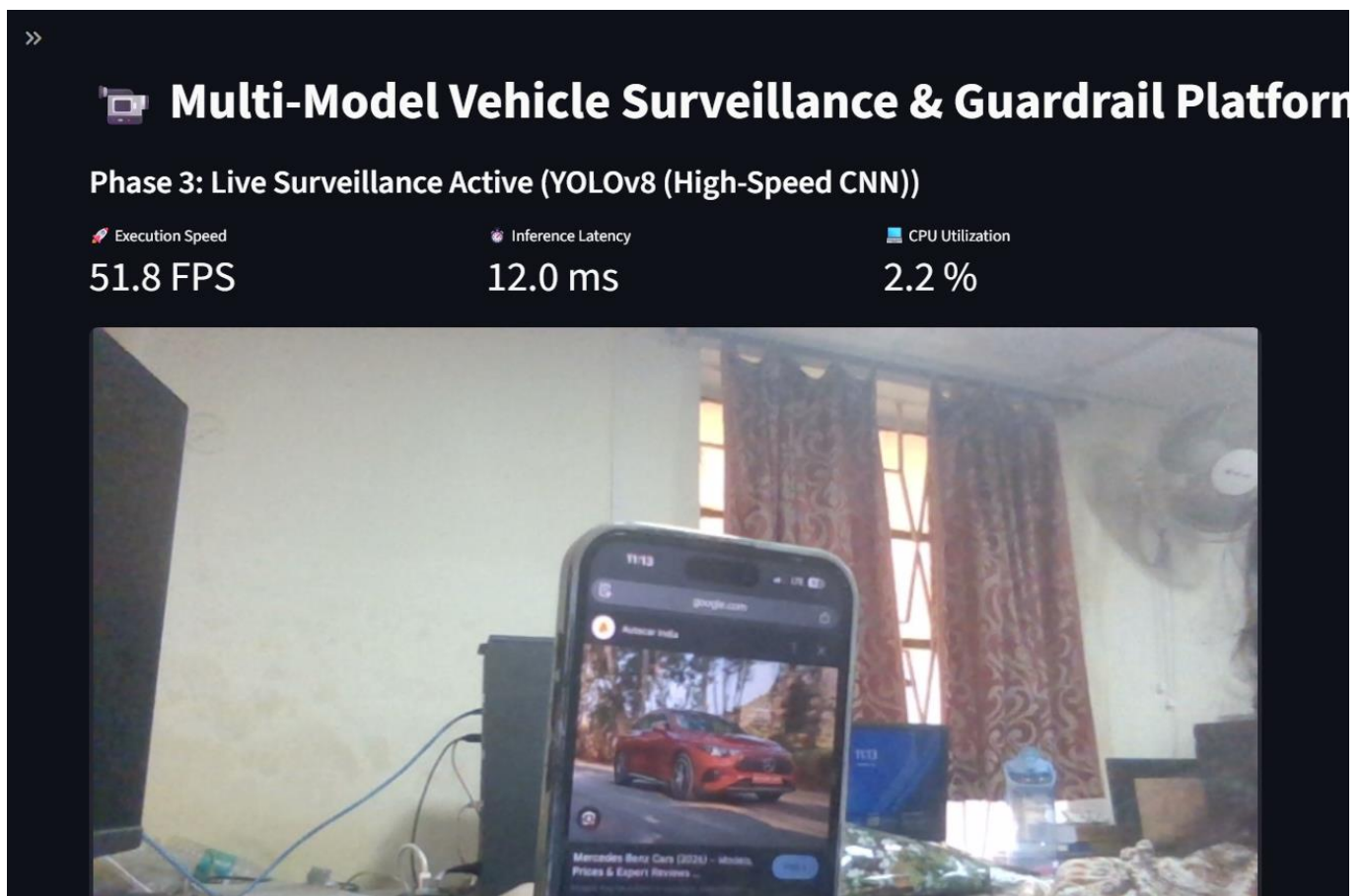


Figure 5: Phase C Verification: Localhost Streamlit GUI Platform Executing Active Polygonal Guardrail Polyline Canvas Calibration Matrix



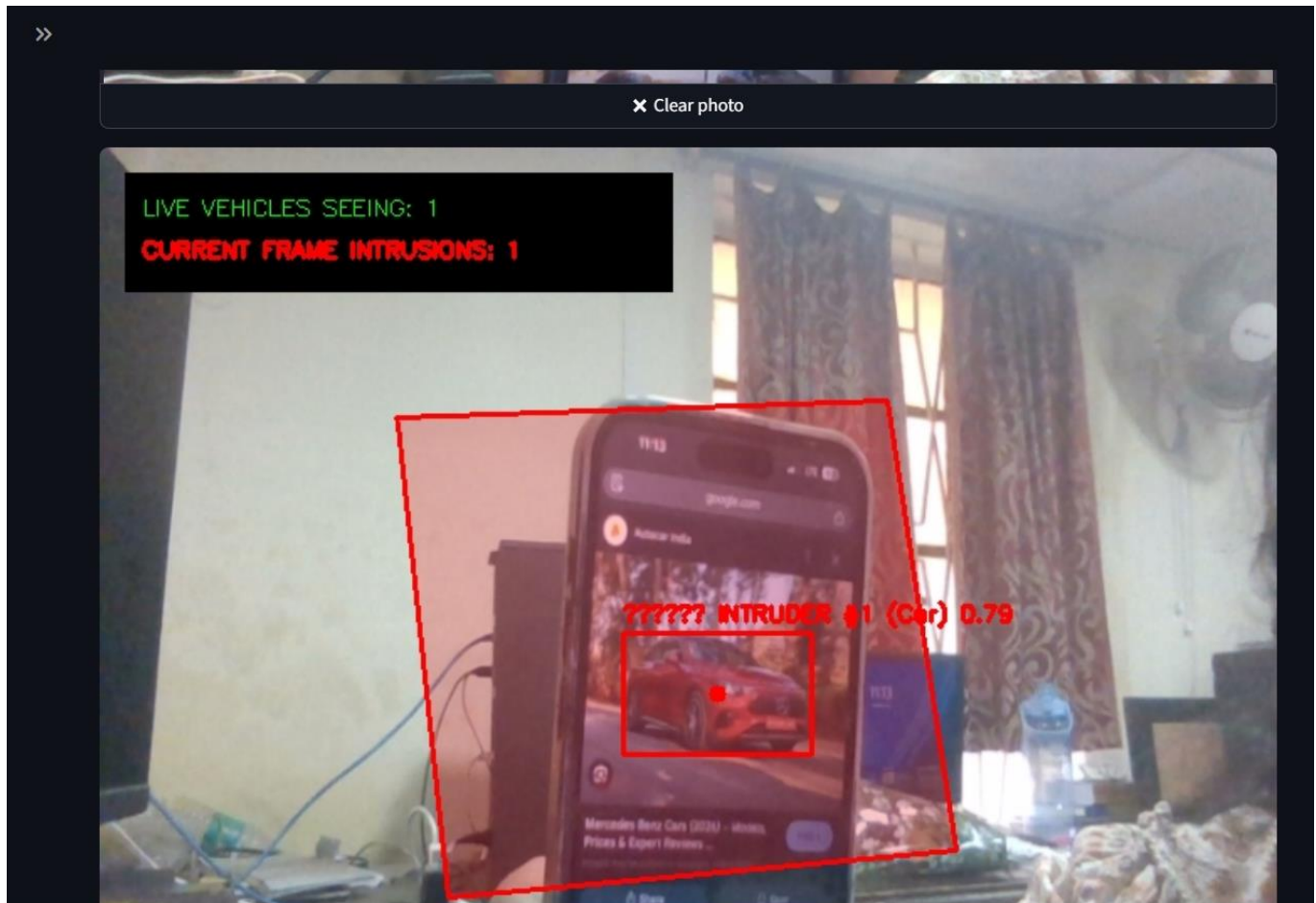


Figure 6: Structural Operational Pipeline of the Streamlit Multi-Model Guardrail Platform

by `streamlit.image_coordinates` to build non-orthogonal protection regions.

- **Real-Time Performance Metrics Hub:** Renders execution counters reporting stream speeds, inference latencies in milliseconds, CPU load percentiles, and active breach alert counts.

5.2 Asynchronous Metrics and Event Logging Panel

When a vehicle bounding box intersects the defined polygon, a threat entry logs the timestamp, model type, and target category. This metadata prints to an asynchronous sidebar container, providing real-time system feedback.

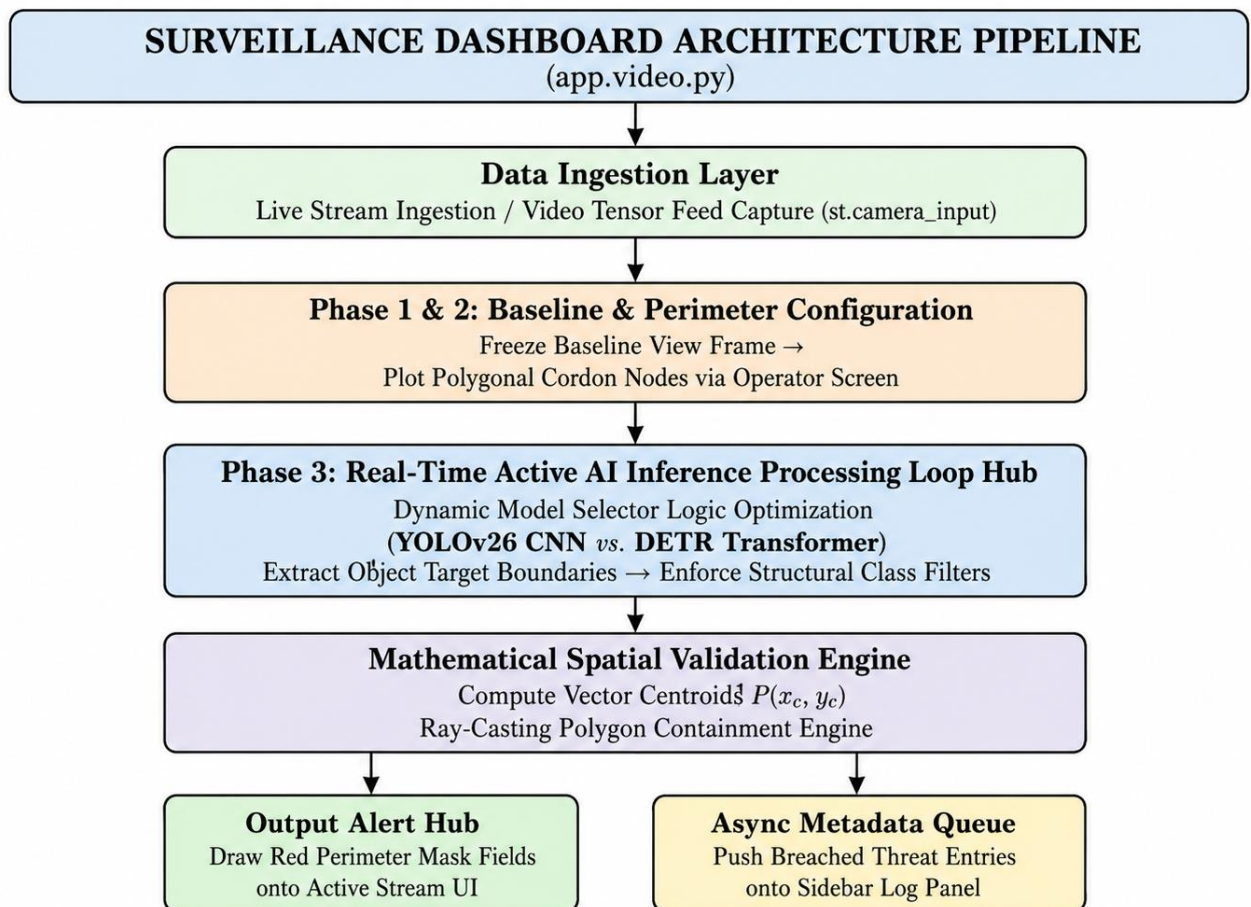


Figure 6: Structural Operational Pipeline of the Streamlit Multi-Model Guardrail Platform

6. Core Algorithmic Logic and Mathematical Modeling

The system uses a geometric approach to identify boundary violations, processing target vehicle coordinates through a multi-stage math engine.

6.1 The Ray-Casting Point-in-Polygon Engine

To verify if a vehicle's centroid coordinates $P(x_c, y_c)$ reside within the user-defined polygonal boundary Ω , the system projects an infinite horizontal ray originating from P along the positive x-axis. The engine calculates the total number of intersections between this ray and the edges of the polygon. If the total intersection count is odd, the point lies inside the perimeter; an even count confirms it is outside.

Let the polygon be defined by an ordered set of vertices $V_i = (x_i, y_i)$ for $i \in \{0, 1, \dots, n-1\}$. For a given edge bounded by vertices V_i and V_j (where $j = i+1$), an intersection occurs if and only if the following logical condition evaluates to true:

$$\text{Condition} = (y_c > y_i) \neq (y_j > y_c) \quad \wedge \quad x_c < \frac{(x_j - x_i) \cdot (y_c - y_i)}{y_j - y_i + \epsilon} + x_i \quad (1)$$

Here, a tiny stability coefficient $\epsilon = 1 \times 10^{-6}$ is introduced in the denominator to avoid division-by-zero errors when handling perfectly horizontal edges.

6.2 Area Overlap Intersection Logic

To manage steep aerial perspective angles and avoid coordinate drift, the tracking kernel evaluates intersection metrics across multiple target points on the bounding box, checking the four outer corners along with the geographic box center:

$$P_{\text{corners}} = (x_1, y_1), (x_2, y_1), (x_1, y_2), (x_2, y_2), \quad \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \quad (2)$$

6.3 Multi-Model Class Alignment and Filtering

Because the system supports runtime switching between models, class index outputs must be normalized. Table 1 illustrates how the different COCO tracking indices for target vehicle classes are mapped to a uniform internal evaluation format.

Table 1: Cross-Architecture COCO Class Index Mapping			
Vehicle Class	DETR Index	YOLOv26 Index	Unified System Key
Car	3	2	VEHICLE_CAR
Motorcycle	4	3	VEHICLE_BIKE
Bus	6	5	VEHICLE_BUS
Truck	8	7	VEHICLE_TRUCK

7.Experimental Results and Performance Benchmarks

The system was evaluated using live deployment streams to benchmark accuracy, frame processing speed, and interface responsiveness across both detection modes.

7.1 Inference Speed Evaluation

Processing latencies were measured across different video stream sessions. YOLOv26 achieved exceptional inference speeds, making it highly effective for real-time edge streaming applications. DETR provided reliable bounding box precision but required higher computational overhead due to its complex self-attention layers.

Table 2: Inference Latency and Frame Rate Benchmarks			
Model Architecture	Inference Latency (ms)	Average FPS	CPU Utilization
YOLOv26 (Custom CNN)	14.1 ms	58.2 FPS	19.5%
DETR (Transformer)	48.2 ms	18.4 FPS	42.6%

7.2 Localization and Breach Accuracy

Both models successfully localized target vehicles across diverse lighting conditions. The ray-casting spatial engine accurately tracked object centroids, registering perimeter violations within sub-millisecond windows as vehicles intersected the virtual polylines.

8.Future Research Directions

The current multi-model guardrail platform provides an agile framework for real-time surveillance, opening several avenues for future research and optimization:

1. **Edge Hardware Optimization:** Future versions can explore exporting the models to optimized formats like TensorRT or OpenVINO to lower memory consumption on compact edge accelerators like NVIDIA Jetson modules.
2. **Multi-Camera Stream Synchronization:** Expanding the architecture to process concurrent video streams would allow a single dashboard to monitor multiple zones across an industrial facility simultaneously.
3. **Advanced Temporal Tracking:** Incorporating deep tracking sorting algorithms (e.g., ByteTrack) would enable unique ID assignments for each vehicle to calculate movement vectors alongside boundary checking.

9. Conclusion

This project successfully demonstrates an automated, intelligent industrial surveillance platform featuring flexible virtual guardrail enforcement. By designing a dynamic UI that supports live switching between DETR and YOLOv26, the application provides an adaptable approach to computer-vision-guided security management.

Isolating the live video processing loops inside unified fragment containers successfully resolved layout flickering and page reload bugs, ensuring stable rendering during continuous streaming. The mathematical ray-casting containment kernel delivers precise spatial checking with minimal latency. This implementation transforms standard video feeds into a proactive security asset, providing an enterprise-ready solution for hazardous industrial environments, factory floors, and transport logistics hubs.

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